Instructor Manual

Competitive Data-Center Connection Planning Game

Power Systems Course

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Instructor purpose. This manual explains how to configure, validate, evaluate, audit, and discuss the competitive data-center connection-planning game. It is the operational companion to the student assignment handout.

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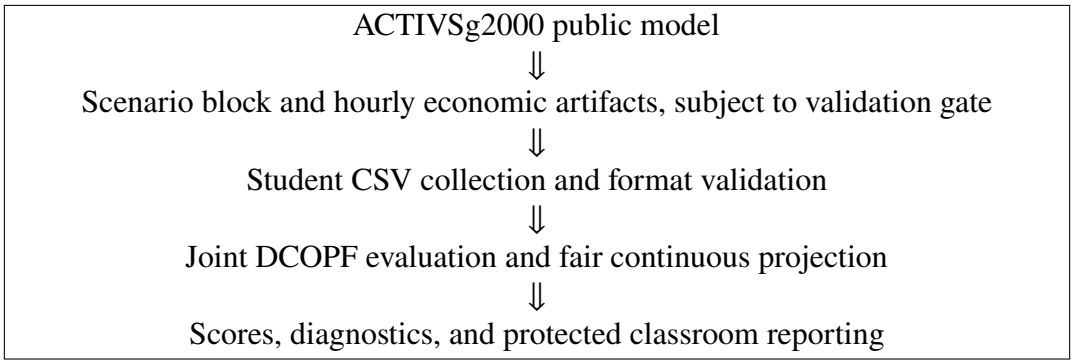
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1 Instructor overview

Each group represents a data-center operator. A group chooses connection sites, site capacities, optional private generation, and optional private storage. The instructor collects one canonical site-level CSV per group and evaluates the submissions jointly with a fixed-load DC optimal power flow (`DCOPF`). Groups are ranked by their final score.

Students are explicitly encouraged to use AI tools, including ChatGPT, Codex, Claude Code, GitHub Copilot, and similar systems. The instructor should grade the submitted artifact and the students' engineering explanation, not whether students manually typed every line of code. Students remain responsible for validating generated code and understanding their plan.

The recommended production workflow is:



2 Production mental model

The release uses the following distinction:

- **Public dataset:** the synthetic Texas ACTIVSg2000 topology, buses, branches, generators, status values, PMIN, PMAX, and marginal costs.
- **Derived scenario:** deterministic bus-level hourly load profiles over six released half-year blocks, grouped into three weekly matches.
- **Instructional assumptions:** student-owned assets, IT value, capital costs, storage costs, DCOPF settlement, scoring, and fair projection.

The active scenario package contains 2,000 buses, 3,206 branches, and 544 generator records. ACTIVSg2000 does not provide technology labels or hourly availability fields. Therefore the active release uses each active generator's dataset status, PMIN, and PMAX in every hour; it does not apply synthetic wind/solar derating. The package is not historical ERCOT telemetry, an official ERCOT planning model, or a forecast.

Team-built resources follow a separate convention. A team's `bus_id` must refer to a public bus in the selected scenario block. The canonical CSV submits generator and storage capacities directly; it does not select an existing public generator or storage device by ID. Every team-built generator uses the same convex piecewise-linear operating curve at every bus: 25% of capacity at \$80/MWh, the next 50% at \$150/MWh, and the final 25% at \$250/MWh. Students submit capacities; the evaluator determines hourly dispatch. Team-built generators and storage serve only their submitting team.

3 Released scenario blocks

Each round covers one half-year with a one-hour time step.

Year	Round	Evaluator hours	CSV rows
2028	January–June	4,368	one row per site
2028	July–December	4,416	one row per site
2027 or 2029	January–June	4,344	one row per site
2027 or 2029	July–December	4,416	one row per site

The classroom schedule has three weekly matches and six submissions:

Week	Submission 1	Submission 2	Ranking and state treatment
1	`2027_jan_jun`	`2027_jul_dec`	Practice week: feedback only; state persists within the week; neither submission counts.
2	`2028_jan_jun`	`2028_jul_dec`	Independent scored match; Submission 1 and Submission 2 both count.
3	`2029_jan_jun`	`2029_jul_dec`	Independent scored match; Submission 1 and Submission 2 both count.

Accepted sites, evaluator-assigned assets, and final storage SOC persist from Submission 1 to Submission 2 within a week. Capital cost (connection-site cost per MW, generator capital, storage capital) is charged once, in the submission where the asset is first built: Submission 1 pays for everything it builds, and Submission 2 pays only for newly added sites and assets, not for those carried from Submission 1. The asset ledger and SOC reset at the beginning of the next week. The final ranking is the sum of four scores: Week 2 Submission 1 + Week 2 Submission 2 + Week 3 Submission 1 + Week 3 Submission 2; Week 1 is excluded and an invalid submission scores 0.

The released artifacts are under `data/processed/activsg2000_scenarios_v1/<year>/<block>`. The root and block manifests record validation status, scenario hashes, and artifact metadata. In this checkout, all six blocks are `validation_status: unverified_dependency` because `highspy` is unavailable. This is a dependency limitation, not a passed or failed validation. A block must report `validation_status: passed` after a complete HiGHS-backed run before it is used for a classroom round.

4 Student submission contract

The active CSV header is exactly seven columns:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
```

The file has one data row per site and no hourly horizon rows. It has no `period`, hourly `load_mw`, `submission_id`, `generator_id`, or `storage_id` columns. `site_id` is a team-defined site label; `bus_id` must come from the selected block's bus catalog. The evaluator uses the submitted site capacities with the released hourly scenario and applies the common team-built generator curve

described above. It assigns deterministic private generator and storage IDs from the team/site identity. Students do not submit those IDs. Generator and storage assets are private to the submitting team and cannot serve another team.

Example:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
team-blue,site-north,1001,40,20,10,40
team-blue,site-south,1002,15,0,0,0
```

The retired ten-column hourly format is historical compatibility input only. Do not distribute it as a current student template.

Students submit storage capacities only. The grid operator schedules storage in the centralized prepared HiGHS model. Storage uses MW power capacity and MWh energy capacity. With ideal charging and discharging efficiency, the evaluator represents each asset by one signed net-power variable per period (positive discharges, negative charges), so the coupled horizon is a linear program with no binary operating-mode variable. It enforces power and energy bounds, discharge no greater than the stored energy, an initial state of charge of $0.5B$, and a terminal state of charge of at least $0.5B$. Discharge can reduce only the owning team's load.

5 Course operation

Before Week 1, provide students with the student handout, the active scenario package, the CSV checker, and the bus catalog. Collect two submissions per week for the three weekly matches. The instructor should record the week, submission number, year/block, deadline, submission directory, rule-file version, and a hash of each frozen input directory.

Recommended sequence:

1. Explain the data/model distinction and the CSV contract.
2. Give groups time to explore candidate buses and resource portfolios.
3. Require each group to run the checker before submission.
4. Freeze the submission directory at the deadline.
5. Run the batch evaluator once for each submission, or run the classroom coordinator for the complete six-submission prefix.
6. Release Week 1 feedback and privacy-safe aggregate/team summaries without exposing competitors' private plans.

6 Instructor command cookbook

Check one submission:

```
ercot-game-check-csv --file path/to/team.csv
```

Evaluate a frozen round:

```
ercot-game-batch run \
  --submissions-dir work/round_01/submissions \
  --artifact-dir data/processed/activsg2000_scenarios_v1/2028/jan_jun \
  --rules data/teaching/dispatch_scoring_rules_jan_jun_v1.json \
  --year 2028 --round jan_jun \
  --submission-number 2 \
  --workers 8 --projection-iterations 10 \
  --output-dir work/round_01/results
```

The submission number is evaluator-side metadata, not a CSV column. The CLI accepts `1` or `2` and defaults explicitly to `2`. Use `--submission-number 1` for a Submission 1 run: it excludes energy-purchase cost while retaining the other active scoring terms. Use `--submission-number 2` for a Submission 2 run, which settles grid purchases at the locational marginal price (LMP) from the selected joint dispatch. In Weeks 2 and 3 both submissions are scored. A standalone `run` has no prior-submission asset ledger and therefore charges capital for every asset in the files; the `classroom` command below carries Submission 1 assets into Submission 2 without a second capital charge.

For an end-to-end classroom run, provide a manifest containing the six released rounds and use:

```
ercot-game-batch classroom \\
  --manifest work/classroom_manifest.json \\
  --state-path work/classroom_state.json \\
  --output-dir work/classroom_results
```

The classroom coordinator rejects a second submission without a committed first submission for that week, persists the asset ledger and SOC within the week, resets them between weeks, charges capital only in the submission where an asset is first built, ranks each scored week by the Submission 1 + Submission 2 total, ranks the final standings by the sum of the four Week 2 and Week 3 scores, and writes aggregate classroom-safe outputs.

Validate the complete scenario package:

```
PYTHONPATH=.:src python3 tools/activsg2000/validate_scenario_package.py \
  --package data/processed/activsg2000_scenarios_v1 --workers 6
```

The economic profiles are generated by the module invocation below. The module form is intentional because the builder uses package-relative imports:

```
PYTHONPATH=.:src python3 -m tools.activsg2000.build_economic_profiles \
  --artifact-dir data/processed/activsg2000_scenarios_v1/2028/jan_jun \
  --output-dir data/processed/activsg2000_scenarios_v1/2028/jan_jun
```

7 Evaluation and joint infeasibility

Malformed files are rejected independently so that valid teams remain eligible for evaluation. A valid plan may be feasible alone but infeasible when all teams are dispatched together. Before projection, the evaluator caps each site at 100 MW, each team at 500 MW, and in multi-team cases caps the

largest request at the smallest positive team request. It then chooses a separate $\alpha_i \in [0, 1]$ for each team: first maximize total served load, then within $\max(10^{-6}, 10^{-6}L^*)$ MWh of the optimum maximize $\sum_i \log(\alpha_i d_i)$. The output records each team's service fraction, projected load, cap policy, projection scope, and proportional-fair utility. Submitted asset costs remain part of the score, while served-load value and energy costs reflect the projected load.

If a projected dispatch cannot be found, the evaluator records deterministic leave-one-team-out and individual fallback attempts where possible. This prevents one invalid or excessively large submission from silently erasing the scores of otherwise valid teams.

8 Scoring model

At a high level, a team's score is:

$$\begin{aligned}&\text{IT value} - \text{energy purchase cost} - \text{generator operating cost} - \text{generator capital cost}\\ &\quad - \text{storage capital cost} - \text{site cost}.\end{aligned}$$

The active instructional coefficients are IT value of \$6500/MWh, a team-built generator curve of \$80/\$150/\$250 per MWh at 25/50/25%, site cost of \$25m/MW, generator capital cost of \$1.5m/MW, and storage capital costs of \$400,000/MW plus \$250,000/MWh. A four-hour storage device therefore has a \$1.4m/MW capital equivalent, and storage operating cost is not charged. The site coefficient is an instructor-chosen mapping informed by JLL shell/core and AI fit-out construction costs [1]; EIA/NREL technology and battery benchmarks inform generator and storage calibration [3, 5, 2, 4].

For Submission 2, energy purchase cost uses the nonnegative LMP from the selected joint dispatch at the selected bus and hour. The released LMP artifacts are deterministic economic-profile references with one row per hour and one column per bus; raw balance duals are retained for audit. Submission 1 has no energy-purchase cost (it is still scored in Weeks 2 and 3). IT value is a complete full-horizon vector supplied in the corresponding economic-profile JSON. These are instructional model outputs, not official ERCOT market prices. The submission number is selected on the instructor CLI and is never represented by an extra CSV column.

9 Current eight-team reproducible example

The repository includes a current-format eight-team demonstration at `docs/instructor_manual/examples/eight_team_round/`. It is a synthetic instructor fixture, not an official student result or a calibration target. Each team has two sites; the first site includes a private generator and four-hour storage, while the second site is connection-only. Two teams request bus `7049` to demonstrate joint competition.

Regenerate and check the eight inputs from the repository root:

```
python3 scripts/simulate_eight_team_round.py
for file in docs/instructor_manual/examples/eight_team_round/submissions/*.csv;
    do
  PYTHONPATH=src python3 scripts/check_submission.py --file "$file"
done
```

The full-horizon instructor command is:

```
PYTHONPATH=src python3 -m ercot_game.batch_cli run \\
  --submissions-dir docs/instructor_manual/examples/eight_team_round/
    submissions \\
  --artifact-dir data/processed/activsg2000_scenarios_v1/2028/jan_jun \\
  --rules data/teaching/dispatch_scoring_rules_jan_jun_v1.json \\
  --horizon 4368 --workers 4 --submission-number 2 \\
  --output-dir work/eight_team_round_results
```

The verified Electrify run accepted all eight files and found a feasible joint dispatch, so every team received its complete requested capacity. It took 1,216.75 seconds (20.3 minutes) and used approximately 9.65 GiB peak resident memory. The compact result is stored in `docs/instructor_manual/examples/eight_team_round/results.json`; large raw evaluation and per-period dispatch files remain outside the repository.

Rank	Team	Buses	Request (MW)	Connected (MW)	Gen (MW)	Storage (MW/MWh)	Score (USD
1	team-08	7201, 7183	55	55	10	5/20	-762,613,56
2	team-07	7248, 5465	65	65	12	6/24	-901,611,60
3	team-06	5084, 7004	70	70	12	6/24	-969,235,85
4	team-05	7039, 4040	80	80	15	8/32	-1,110,683,93
5	team-04	6199, 5103	90	90	20	10/40	-1,254,731,50
6	team-03	7229, 5385	100	100	18	8/32	-1,385,079,76
7	team-02	7049, 5449	115	115	40	12/48	-1,621,150,91
8	team-01	7049, 4071	130	130	24	12/48	-1,803,224,32

The total connected capacity is 705 MW and total served load is 3,079,440 MWh. This recorded run predates the current coefficients: it used the superseded pilot values (IT value $1,500/MWh, site cost $20m/MW, generator capital $1.3m/MW, storage capital $350,000/MW plus $200,000/MWh), so its scores do not reflect the active rule files. Scores are negative because site cost dominated the IT-service value under those pilot values. It is also a standalone Submission 2 run, so every asset is charged capital. The private generators are present in the plans but were not dispatched in this joint optimum, so their operating-cost component is zero. The complete score decomposition is in `results.json`.

10 Concrete multi-team example

The complete example package is in the following historical compatibility directory:

```
docs/instructor_manual/examples/multi_team_round/
```

It is retained as a historical long-form compatibility package with matching results. It is not a current student CSV template; current submissions use the canonical one-row-per-site contract above. The full raw evaluation record is not checked into the repository because it is very large.

10.1 Single-team CSV excerpt

Each student group submits one CSV file. The current format contains one row per site and no hourly horizon rows:

```
team_id,site_id,bus_id,connection_capacity_mw,generator_capacity_mw,
    storage_power_mw,storage_energy_mwh
team-blue,site-north,1001,40,20,10,40
team-blue,site-south,1002,15,0,0,0
```

The public `bus_id` values must come from the current 2,000-bus artifact; `site_id` is a team-defined label. The checked-in CSVs in the historical example directory retain the retired hourly format solely to preserve a reproducible historical result and must be labeled legacy whenever shown.

10.2 Batch collection example

For a round with four groups, the instructor collects four separate files, for example `team-01.csv`, `team-02.csv`, `team-03.csv`, and `team-04.csv`. The batch evaluator reads the directory as a joint submission, but no CSV should contain rows for multiple teams. The leaderboard below is therefore a multi-team result report, not a multi-team submission file.

10.3 Example score report

The following table is populated from the exact generated leaderboard and is not a hand-selected score illustration. It is a historical/legacy compatibility result: its checked-in rules, values, and cost fields predate the active implementation and must not be used as a calibration target for a new round.

Rank	Team	Served load (MWh)	IT value ($)	Energy cost ($)	Asset/storage cost ($)	Score ($)
1	team-03	41,318.88	5,164,859.70	724,983.76	50,000.00	4,389,875.94
2	team-01	41,599.96	5,199,994.80	467,665.19	567,465.38	4,164,864.23
3	team-04	26,779.33	3,347,416.80	301,047.10	388,716.50	2,657,653.19
4	team-02	40,279.73	5,034,966.30	706,971.84	1,945,424.52	2,382,569.94

Asset/storage cost combines site cost, generator capital cost, generator operating cost, storage capital cost, and storage operating cost. Exact component fields are preserved in `leaderboard.csv`; all four submissions were valid and feasible in the joint run.

The decomposition should be read together with the compact `evaluation_summary.json` and the exact `leaderboard.csv`. In a live instructor run, retain the complete raw evaluation record separately. In particular, a large served-load value can improve IT revenue while increasing locational energy cost; storage can reduce or shift energy purchases but adds power-capacity and energy-capacity costs.

10.4 Artifacts to inspect

- `leaderboard.csv`: rank, feasibility, score, and major cost/value fields;
- `submission_manifest.csv`: input validity and SHA-256 hashes;

- `evaluation_summary.json`: compact run metadata, accepted teams, projection outcome, and per-team scores;
- `diagnostic_team_01.json`: one representative team-level diagnostic record;
- `dispatch_period_0.json`: one representative dispatch record;
- `rules/dispatch_scoring_rules_jan_jun_v1.json`: the exact rule file used by the run;
- `evaluation_sha256.txt`: hash of the evaluation record.

11 Privacy and classroom reporting

Keep raw CSVs, per-team diagnostics, bus selections, generator capacities, storage capacities, and hourly profiles instructor-only. The classroom writer's shared `classroom-safe-v1` output may include aggregate team counts, aggregate periods, and aggregate projection metadata. Each team receives only its own `classroom-private-v1` summary under a hashed directory, with its own score and scalar diagnostics. Never publish a map or raw JSON that lets one group reconstruct another group's exact sites or resource portfolio.

12 Reproducibility and release audit

Before release, verify:

- all six scenario blocks report a passed validation manifest; if `highspy` is unavailable, record `unverified_dependency` and do not release the block;
- every economic-profile NPZ hash matches its manifest;
- the chosen rule file and scenario block match the round announcement;
- a dry-run CSV passes the checker and batch evaluator;
- the output directory contains the leaderboard, manifest, evaluation record, diagnostics, and hashes;
- the frozen submission directory is archived before any feedback is released.
- the Colab benchmark has a completed full-horizon result before making any performance or horizon-reduction claim; the benchmark requires `highspy`, `psutil`, prepared artifacts, and a compatible runtime.

13 Troubleshooting

Malformed CSV

Run the checker and return the error to the CSV-format section of the student handout. Do not repair a student's file silently.

Unknown bus
Confirm that the `bus_id` comes from the selected half-year artifact, not from a retired prototype.

Duplicate site
Ensure that every canonical data row has a distinct `site_id`.

Joint infeasibility
Inspect projection metadata and binding diagnostics. A valid team may still receive a projected score.

Missing economic profile
Confirm that the selected block contains all of the following files and that the manifest hash matches:

- `economic_profiles.npz`
- `economic_profile.json`
- the block manifest

14 Instructor checklist

1. Select the three-week schedule, submission number, rule file, and deadlines.
2. Publish the student handout and checker.
3. Freeze and hash the submission directory.
4. Run format validation and the full batch evaluation for each submission, or run the classroom coordinator.
5. Confirm weekly persistence, capital charged only in the submission where an asset is first built, and reset at the start of Weeks 2 and 3.
6. Inspect valid/invalid counts, per-team projection scope, and diagnostics.
7. Release protected summaries and retain the complete instructor archive.

References

[1] Jones Lang LaSalle IP, Inc. 2026 global data center outlook, 2026.

[2] National Renewable Energy Laboratory. Annual technology baseline 2024: Utility-scale battery storage. Technical report, 2024.

[3] U.S. Energy Information Administration. Capital cost and performance characteristics for utility-scale electric power generating technologies, 2024.

[4] U.S. Energy Information Administration. Capital cost estimates for utility scale electricity generating plants: Battery energy storage system assumptions, 2024.

[5] U.S. Energy Information Administration. How much electricity is generated by natural gas-fired power plants?, 2025.